



Answers & Solutions:

1. **A. 1300 nm, 1550 nm and 850 nm**
2. **B. can be caused by microbends and macrobends**
3. **C. is the lengthening of light pulses as they travel down the fiber**
4. **C. passes repeatedly through the center of the core**
5. **A. is used to prevent accidental exposure to invisible light**
6. **C. is a plastic layer around the cladding to protect the optic fiber from mechanical damage**

7. **A. 2.2 μm**

Solution:

$$r_{\max} = \frac{0.383\lambda}{\text{NA}}$$

$$r_{\max} = \frac{0.383(1 \times 10^{-6})}{0.35}$$

$$r_{\max} = 1.09 \mu\text{m}$$

$$d_{\max} = 2r_{\max}$$

$$d_{\max} = 2(1.09 \times 10^{-6})$$

$$d_{\max} = 2.2 \mu\text{m} \rightarrow \text{Ans.}$$

8. **C. an optic fiber**
9. **D. impurities in the fiber**
10. **D. A ray that propagates through the fiber by total internal reflection**
11. **C. Transmits and diffuses them**
12. **B. 34.61°**

Solution:

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$\theta_2 = \sin^{-1} \left(\frac{n_1}{n_2} \sin \theta_1 \right)$$

$$\theta_2 = \sin^{-1} \left[\left(\frac{1.3}{1.28} \right) \sin 34^\circ \right]$$

$$\theta_2 = 34.61^\circ \rightarrow \text{Ans.}$$

13. **D. 1.55 μm**
 14. **D. Splices, couplers, and connectors**
 15. **D. 0.9**
 16. **D. a hybrid cable**
 17. **B. is also called chromatic dispersion**
 18. **A. a light source, an optic fiber and a photo-electric cell**
 19. **B. TIR will always occur**
 20. **A. 18**
 21. **A. 54.34°**
- Solution:
- $$\theta_{\text{in}} = \sin^{-1} \sqrt{n_1^2 - n_2^2}$$
- $$\theta_{\text{in}} = \sin^{-1} \sqrt{(1.55)^2 - (1.32)^2}$$
- $$\theta_{\text{in}} = 54.34^\circ \rightarrow \text{Ans.}$$
22. **A. increase**
 23. **D. plastic fiber is normally used for long distance communications**
 24. **A. is at its highest value at the center of the core**
 25. **B. is never greater than the speed of light in free space**

END